MANCHESTER 1824

The University of Manchester

EART11200 - The Natural Scientist's Toolkit

Mathematics and Physics

Week 8 Practical - Integration By Substitution (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Explain the concept of integration by substitution and its relationship to the chain rule in differentiation.

(2) Apply integration by substitution on indefinite integrals to solve theoretical and real-world problems.

(3) Apply integration by substitution on definite integrals to solve theoretical and real-world problems.

Exercise 1: Integration by Substitution

- Use Integration by Substitution on the following integral:

$$\int \frac{1}{(3x+4)^2} dx$$

Solution 1: Integration by Substitution

1. Write out the integral in its simplest form:

$$y = \frac{1}{(3x+4)^2} dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x + 4$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u - 4)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

Solution 1: Integration by Substitution

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. Make the substitution in the original integral

$$y = \int \frac{1}{(3x+2)^2} dx \implies y = \int \frac{1}{u^2} \frac{du}{3} \implies y = \frac{1}{3} \int u^{-2}\, du$$

7. Integrate as usual:

$$y = \frac{1}{3} \int u^{-2}\, du \implies -\frac{1}{3} u^{-1} + d \implies -\frac{1}{3u} + d$$

8. Substitute the original expression back in to get the final expression:

$$y = -\frac{1}{3(3x+4)} + d$$

Exercise 2: (Definite) Integration by Substitution

- Use Integration by Substitution on the following integral:

$$\int_0^{\pi/2} \sin(2x+1)dx$$

Solution 2: (Definite) Integration by Substitution

1. Write out the integral in its simplest form:

$$y = \int_0^{\pi/2} \sin(2x+1)dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 2x + 1$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{2}(u-1)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{2}$$

Solution 2: (Definite) Integration by Substitution

5. Get an expression for dx:

$$dx = \frac{du}{2}$$

6. Find the new limits of the integral:

$$\text{When } x = 0, \quad u = 2 \times 0 + 1 = 1$$
$$\text{When } x = \frac{\pi}{2}, \quad u = \pi + 1$$

7. Make the substitution in the original integral

$$y = \int_0^{\pi} \sin(2x+1)dx \implies y = \int_1^{\pi+1} \sin(u)\frac{du}{2} \implies y = \frac{1}{2}\int_1^{\pi+1} \sin(u)\, du$$

8. Integrate as usual:

$$\frac{1}{2}\int_1^{\pi+1} \sin(u)\, du = \frac{1}{2}\left[-\cos(u)\right]_1^{\pi+1} = \frac{1}{2}(-\cos(\pi+1) + \cos(1)) = 0.5403$$

Intended Learning Outcomes

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(3) Apply integration by substitution on definite integrals to solve theoretical and real-world problems.